Introduction to CFD (4RC30)

Turbulent flows

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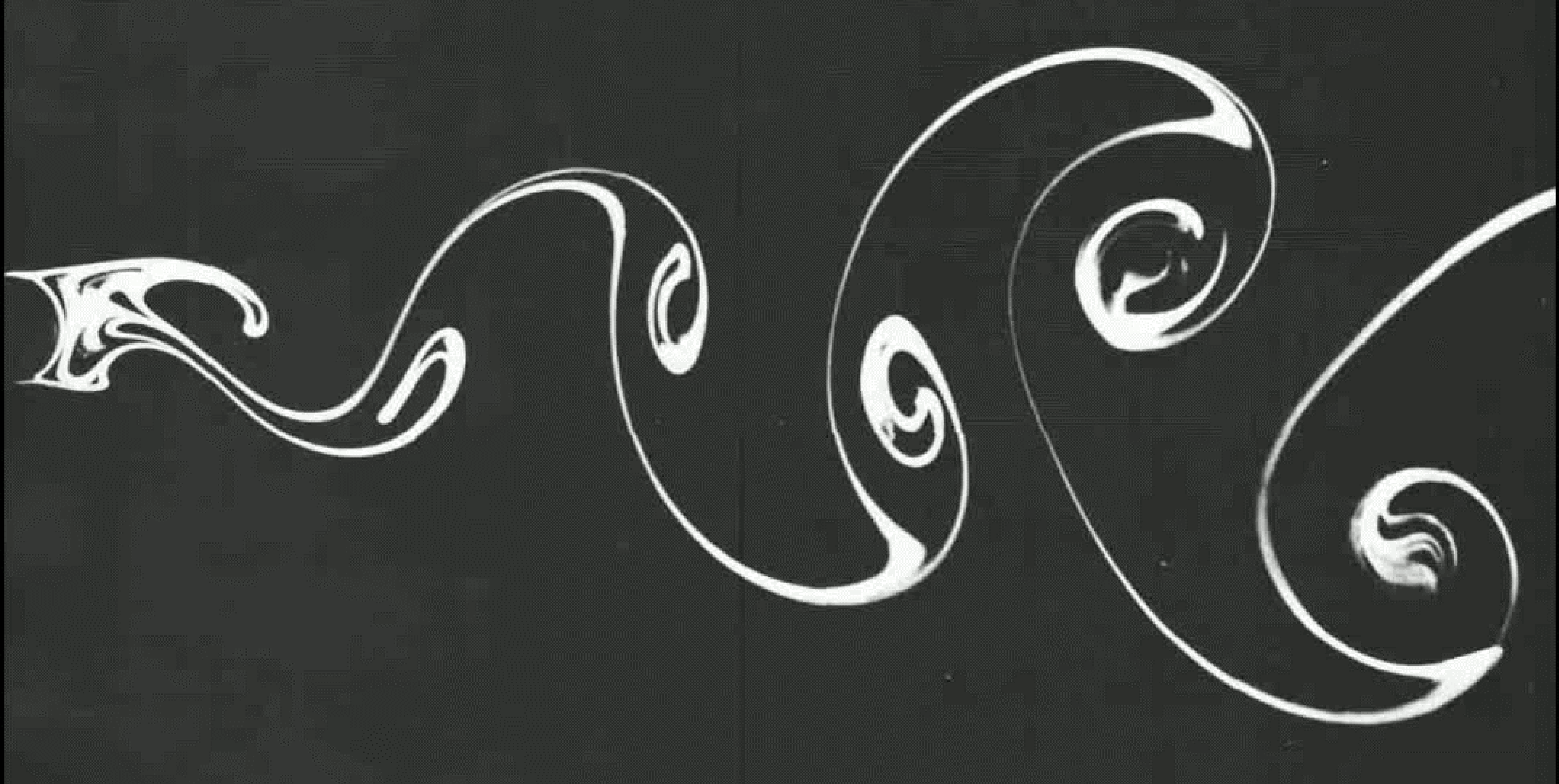
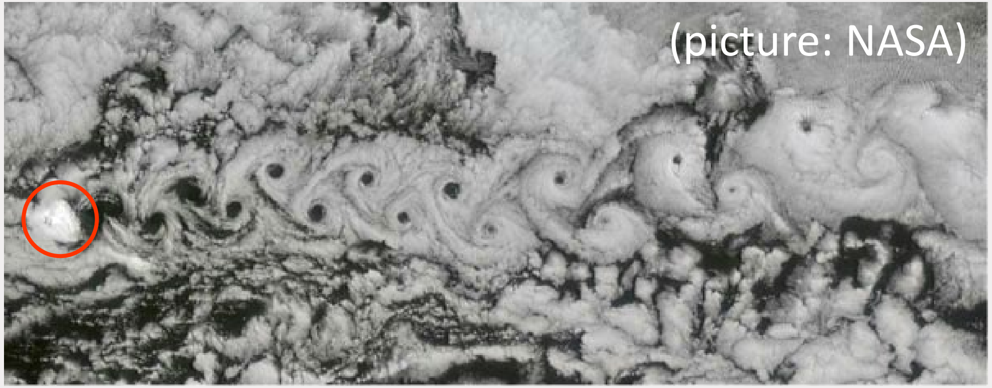
Last week: results vortex shedding



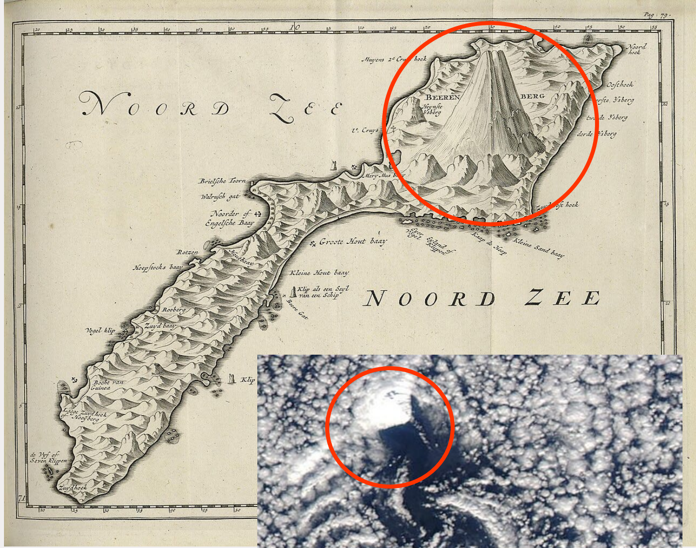
Video credit: https://youtu.be/3mULL6O6f38?si=49NQMJvmXV3_6dFp

Von Karman vortex street

183 km



158 km



Beerenberg volcano
Jan Mayen Island
Height 2.2 km

Image credit: Wikipedia, NASA, Google Maps

Outline

Content of this lecture (Chapter 3)

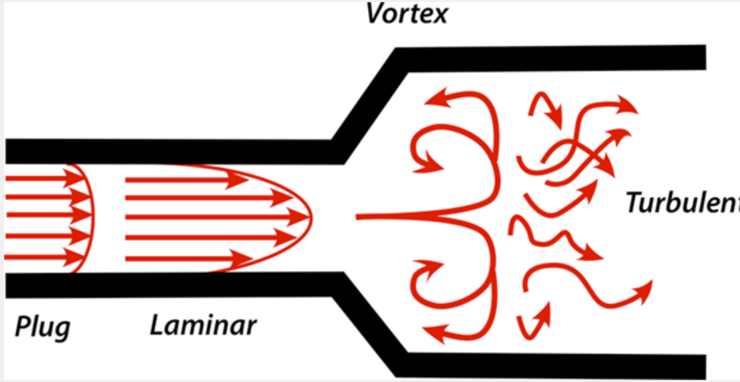
- Qualitative description of turbulence
- Reynolds averaging
- Prandtl mixing length model
- k-ε model
- Large eddy simulations (LES)

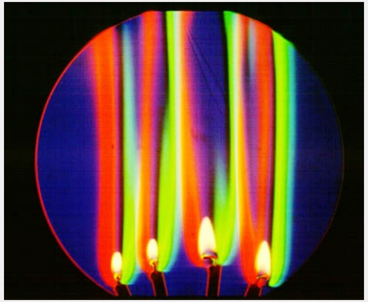
Wrap up

What is turbulence?

- **Definition 1:**
  - Laminar flow: flow in ‘lamina’
  - Turbulent flow: flow with ‘eddies’
  - Problem: laminar vortex shedding
- **Definition 2:**
  - Laminar flow: dampens instabilities
  - Turbulent flow: enhances instabilities
  - Characterized by:

$$\mathrm{Re} = \frac{\text{Convective forces}}{\text{Viscous forces}}$$





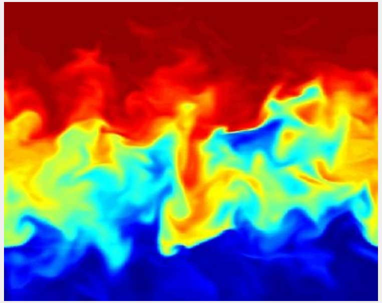


Image credit: https://mriquestions.com/laminar-v-turbulent.html
https://commons.wikimedia.org/wiki/File:Convecting-candles-.jpg
https://thiele.au.dk/fileadmin/www.thiele.au.dk/Events/2015/Aarhus/Birnir.pdf

Role of Reynolds number

- **Re < 2000: laminar flow**
  viscous forces dampen instabilities

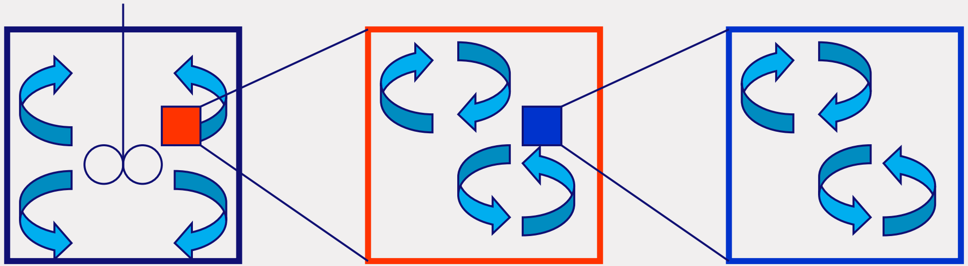
- **For Re > 2300: turbulent flow**
  convective forces enhance instabilities

- **Transition starts at Re ~ 2000 differing from problem to problem!**

Turbulence scales

- **Scale of interest**

Production of energy → Transport of energy → Viscous energy dissipation

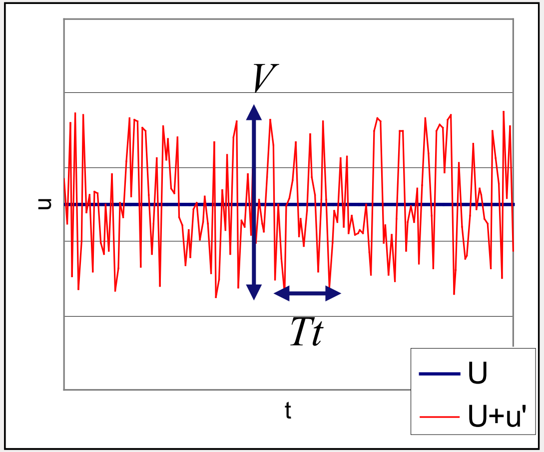


O(m)	O(cm-mm)	O(μm)
Macroscopic models	Typical CFD	DNS
1 cell	(10^2-10^3) cells	10^6 cells

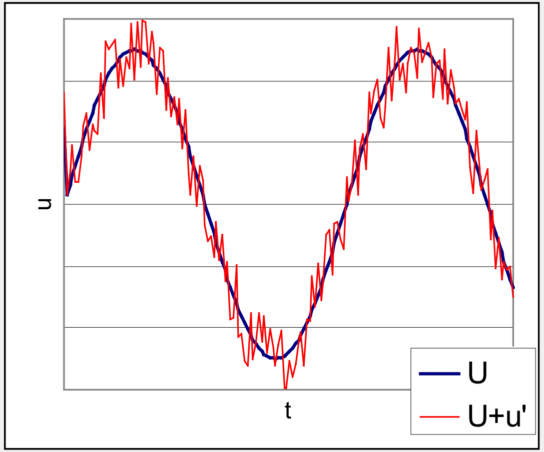
Reynolds averaging

- ***Solve*** **the large scales and *model* the small scales** $u = U + u'$

  - **Steady problems:**
    - Time averaging

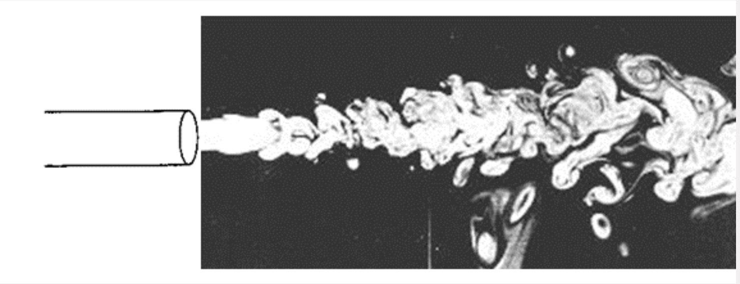


  - **Unsteady problems:**
    - Ensemble averaging



Examples

- **Instantaneous flow field**



- **Reynolds <u>averaged</u> flow field**

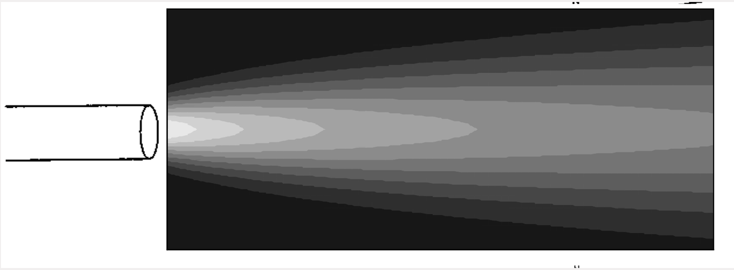


Image credit: https://mriquestions.com/laminar-v-turbulent.html

Reynolds averaging equations

- **Decompose variables in mean and fluctuations:**

$$\boxed{u = U + u'; \qquad v = V + v'; \qquad w = W + w'; \qquad p = P + p'}$$

- **Definition of mean and fluctuation velocities:**

$$U = \frac{1}{\Delta t}\int_0^{\Delta t} u(t)dt \qquad \overline{u'} = \frac{1}{\Delta t}\int_0^{\Delta t} u'(t)dt \equiv 0$$

$$u_{\text{rms}} = \sqrt{\overline{u'^2}} = \left[\frac{1}{\Delta t}\int_0^{\Delta t} (u')^2 dt\right]^{1/2} \geq 0 \qquad k = \tfrac{1}{2}\left(\overline{u'^2} + \overline{v'^2} + \overline{w'^2}\right)$$

Reynolds averaging equations

$$\frac{\partial u}{\partial t} + \mathrm{div}(u\mathbf{u}) = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\,\mathrm{div}(\mathrm{grad}\,u)$$

- **Average the x momentum equation using:**

$$\overline{(u\mathbf{u})} = (\overline{u\mathbf{u}}) = (U\mathbf{U}) + \left(\overline{u'\mathbf{u}'}\right)$$

$$\overline{U} = U, \overline{\mathbf{U}} = \mathbf{U}, \overline{P} = P$$

$$\frac{\partial U}{\partial t} + \frac{\partial \overline{u'}}{\partial t} + \mathrm{div}(U\mathbf{U}) + \mathrm{div}(\overline{u'\mathbf{u}'}) =$$

$$-\frac{1}{\rho}\frac{\partial P}{\partial x} - \frac{1}{\rho}\frac{\partial \overline{p'}}{\partial x} + \nu\,\mathrm{div}(\mathrm{grad}\,U) + \nu\,\mathrm{div}(\mathrm{grad}\,\overline{u'})$$

- **Average of fluctuations are zero, hence:**

$$\overline{u'} = 0, \overline{\mathbf{u}'} = 0, \overline{p'} = 0$$

$$\frac{\partial U}{\partial t} + \mathrm{div}(U\mathbf{U}) + \color{red}{\mathrm{div}(\overline{u'\mathbf{u}'})} = -\frac{1}{\rho}\frac{\partial P}{\partial x} + \nu\,\mathrm{div}(\mathrm{grad}\,U)$$

Reynolds averaging equations

$$\frac{\partial U}{\partial t} + \mathrm{div}(U\mathbf{U}) + \mathrm{div}(\overline{u'\mathbf{u}'}) = -\frac{1}{\rho}\frac{\partial P}{\partial x} + \nu\,\mathrm{div}(\mathrm{grad}\,U)$$

- **Convective momentum transport due to velocity fluctuations:**

$$\mathrm{div}(\rho\overline{u'\mathbf{u}'}) = \frac{\partial(\rho\overline{u'^2})}{\partial x} + \frac{\partial(\rho\overline{u'v'})}{\partial y} + \frac{\partial(\rho\overline{u'w'})}{\partial z}$$

- **Definition of 'Reynolds stresses':**

$$\tau_{xx} = -\rho\overline{u'^2};\quad \tau_{xy} = -\rho\overline{u'v'};\quad \tau_{xz} = -\rho\overline{u'w'} \quad\Rightarrow \boxed{\tau_{ij} = -\rho\overline{u_i'u_j'}}$$

Closures for Reynolds stresses

- **Boussinesq approximation (1877): turbulent fluctuations ~ shear stress**

$$\tau_{ij} = -\rho \overline{u_i ' u_j '} = \mu_t \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right) + \frac{2}{3} \rho k \delta_{ij} \qquad \delta_{ij} = \begin{cases} 1, i = j \\ 0, i \neq j \end{cases}$$

- **Closure models for μ_t:**
  - Zero equation models: Prandtl mixing length
  - Two-equation models: *k*-*ε* model
  - Reynolds stress model and algebraic stress model (see book)

Mixing length model

$$\tau_{ij} = -\rho \overline{u_i' u_j'} = \mu_t \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right)$$

- **Dimensional analysis yields:**

  dynamic viscosity = density · velocity · length;

$$\mu_t = C \rho \mathcal{V} \mathcal{L}$$

$$\mathcal{V} = \mathcal{L} \left| \frac{\partial U}{\partial y} \right| \quad \Rightarrow \quad \boxed{\mu_t = \rho l_m^2 \left| \frac{\partial U}{\partial y} \right|}$$

$$\Rightarrow \quad \tau_{xy} = \tau_{yx} = -\rho \overline{u_x' u_y'} = \rho l_m^2 \left| \frac{\partial U}{\partial y} \right| \frac{\partial U}{\partial y}$$

Assessment mixing length model

- **Assumption made:**

  Turbulence is characterized by *one* velocity scale and *one* length scale, that need to be known a priori (see book for table)

- **Easy to implement and cheap in calculation**

- **Fails for complex flows** (e.g. with separation and circulation)

k-*ε* model

$$\tau_{ij} = -\rho\overline{u_i'u_j'} = \mu_t\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right) - \frac{2}{3}\rho k\delta_{ij} = 2\mu_t E_{ij} - \frac{2}{3}\rho k\delta_{ij}$$

- **Two-equation model based on:**
  - Turbulent kinetic energy: $k = \tfrac{1}{2}\left(\overline{u'^2} + \overline{v'^2} + \overline{w'^2}\right)$ [m²/s²]
  - Viscous dissipation of turbulent kinetic energy: ε [m²/s³]

$$\mathcal{V} = k^{1/2};\ \mathcal{L} = \frac{k^{3/2}}{\varepsilon} \quad \Rightarrow \quad \boxed{\mu_t = C\rho\mathcal{V}\mathcal{L} = \rho C_\mu \frac{k^2}{\varepsilon}}$$

k equation

$$k = \tfrac{1}{2}\left(\overline{u'^2} + \overline{v'^2} + \overline{w'^2}\right) \qquad u_i' = u_i - U_i$$

- **Multiply Navier-Stokes equations with u_i': $u_i\, u_i'$**
- **Multiply Reynolds equations with u_i': $U_i\, u_i'$**
- **Calculate $\Sigma\, u_i u_i'$ - $\Sigma\, U_i u_i'$ to obtain:**

$$\boxed{\frac{\partial(\rho k)}{\partial t} + \mathrm{div}(\rho k\mathbf{u}) = \mathrm{div}\left(\frac{\mu_t}{\sigma_k}\,\mathrm{grad}\,k\right) + 2\mu_t E_{ij}\cdot E_{ij} - \rho\varepsilon}$$

accu-mulation	convective transport	diffusive transport	production	dissipation

ε equation

- **The production and dissipation of turbulent kinetic energy are closely linked:**

  The source and sink terms in the ε equation are proportional to those in the k equation

$$\boxed{\frac{\partial(\rho k)}{\partial t} + \operatorname{div}(\rho k \mathbf{u}) = \operatorname{div}\left(\frac{\mu_t}{\sigma_k} \operatorname{grad} k\right) + 2\mu_t E_{ij} \cdot E_{ij} - \rho\varepsilon}$$

$$\boxed{\frac{\partial(\rho \varepsilon)}{\partial t} + \operatorname{div}(\rho \varepsilon \mathbf{u}) = \operatorname{div}\left(\frac{\mu_t}{\sigma_\varepsilon} \operatorname{grad} \varepsilon\right) + C_{1\varepsilon} \frac{\varepsilon}{k} 2\mu_t E_{ij} \cdot E_{ij} - C_{2\varepsilon} \rho \frac{\varepsilon^2}{k}}$$

$$\boxed{C_\mu = 0.09; \quad \sigma_k = 1.00; \quad \sigma_\varepsilon = 1.30; \quad C_{1\varepsilon} = 1.44; \quad C_{2\varepsilon} = 1.92}$$

Implementation k-ε model in CFD codes

Source term linearization (remember: $\bar{S}\Delta V = S_u + S_P\phi_P$ with $S_u > 0$, $S_P < 0$)

$$\frac{\partial(\rho k)}{\partial t} + Conv = Diff + \underbrace{2\mu_t E_{ij}\cdot E_{ij} - \rho\varepsilon}_{\bar{S}\Delta V}$$

$$\frac{\partial(\rho \varepsilon)}{\partial t} + Conv = Diff + \underbrace{C_{1\varepsilon}\frac{\varepsilon}{k}2\mu_t E_{ij}\cdot E_{ij} - C_{2\varepsilon}\rho\frac{\varepsilon^2}{k}}_{\bar{S}\Delta V}$$

$$S_u = 2\mu_t E_{ij}\cdot E_{ij} \qquad S_p = -\rho\left(\frac{\varepsilon}{k}\right)^{old}$$

$$S_u = C_{1\varepsilon}\left(\frac{\varepsilon}{k}\right)^{old} 2\mu_t E_{ij}\cdot E_{ij} \qquad S_p = -C_{2\varepsilon}\rho\left(\frac{\varepsilon}{k}\right)^{old}$$

Discretization of convection and diffusion using standard form:

$$a_P\phi_P = \sum a_{nb}\phi_{nb} + S_u \qquad a_P = \sum a_{nb} + \Delta F - S_P$$

Assessment *k*-*ε* model

- Simplest turbulence model without a priori knowledge of velocity and length scales
- Successfully and widely applied in industry
- Most widely validated turbulence model
- More expensive than mixing length model (2 extra PDE's)
- Isotropic turbulence assumption not valid for swirling flows, buoyant flows, etc.

Implementation momentum equations

$$\frac{\partial(\rho u)}{\partial t} + \operatorname{div}(\rho u \mathbf{u}) = -\frac{\partial p}{\partial x} + \operatorname{div}(\mu \operatorname{grad} u)$$

Reynolds averaging gives:

$$\frac{\partial(\rho U)}{\partial t} + \operatorname{div}(\rho U \mathbf{U}) = -\frac{\partial P}{\partial x} + \operatorname{div}(\mu \operatorname{grad} U) - \operatorname{div}(\rho \overline{u' \mathbf{u}'})$$

$$\frac{\partial(\rho U)}{\partial t} + \operatorname{div}(\rho U \mathbf{U}) = -\frac{\partial P}{\partial x} + \operatorname{div}(\mu \operatorname{grad} U) + \underbrace{\operatorname{div}(\mu_t \operatorname{grad} U - \tfrac{2}{3} \rho k \delta_{ij})}_{\text{Reynolds stresses: modelled}}$$

$$\frac{\partial(\rho U)}{\partial t} + \operatorname{div}(\rho U \mathbf{U}) = -\frac{\partial P}{\partial x} + \operatorname{div}(\mu_{eff} \operatorname{grad} U - \tfrac{2}{3} \rho k \delta_{ij})$$

$$\mu_{eff} = \mu + \mu_t = \mu + \rho C_\mu \frac{k^2}{\varepsilon}$$

Large Eddy Simulations (LES)

- **Navier-Stokes equations are filtered:** $u = U + u'$

- **Resolve the flow on grid scales (the large eddies) and model the flow on sub-grid scales**

$$\frac{\partial(\rho U)}{\partial t} + \mathrm{div}(\rho U\mathbf{U}) = -\frac{\partial P}{\partial x} + \mathrm{div}(\mu_{eff}\,\mathrm{grad}\,U - \tfrac{2}{3}\rho k\delta_{ij})$$

- **Smagorinsky model (1963)**

$$\mu_{eff} = \mu + \mu_t = \mu + \rho l_m^2 \sqrt{E_{ij} \cdot E_{ij}}$$

$$l_m^2 = (C_S\Delta)^2; \quad \Delta = (\Delta x \Delta y \Delta z)^{1/3}; \quad C_S = 0.08 - 0.2$$

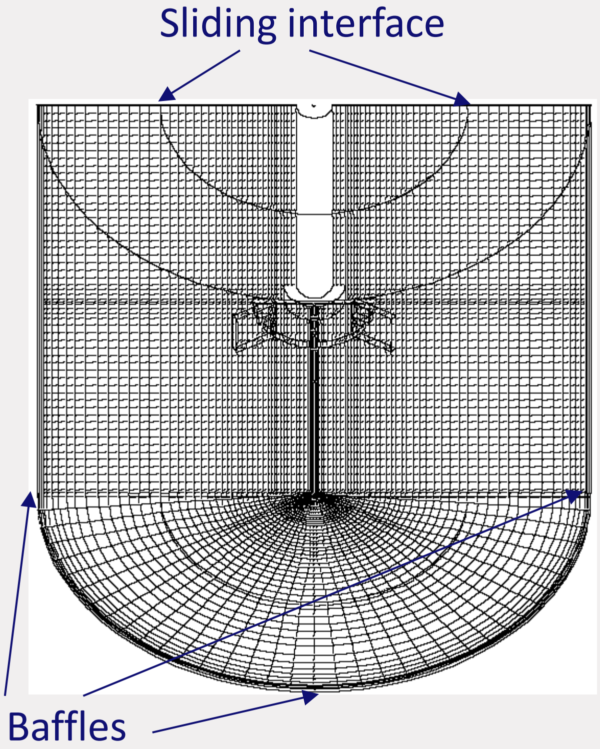
Assessment LES

- Zero-equation SGS model easy to implement
- Captures dynamics of the flow
- Fine grid needed, so very expensive
- Probably best suited for chemically reacting flows and turbulent multiphase flows

Example: stirred tank

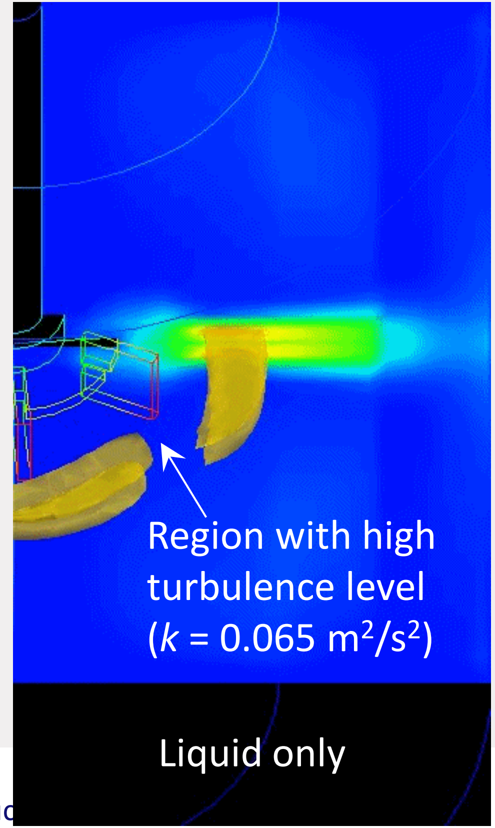
- **Sliding grid**
- **180° geometry**
- **CFX 4.3**
- ***k*-ε model**

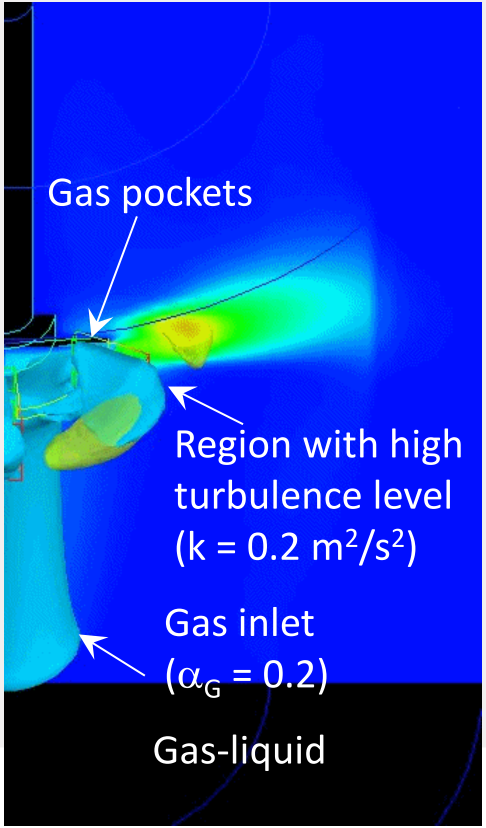
	N_r	N_z	N_q	N_{tot}
Coarse	46	56	36	10^5
Fine	92	112	36	$4 \cdot 10^5$



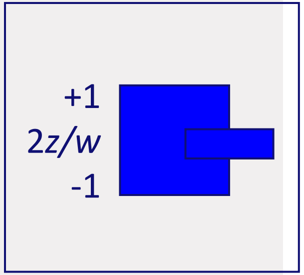
Deen et al. (2002), Can. J. Chem. Eng. 80, 1-15

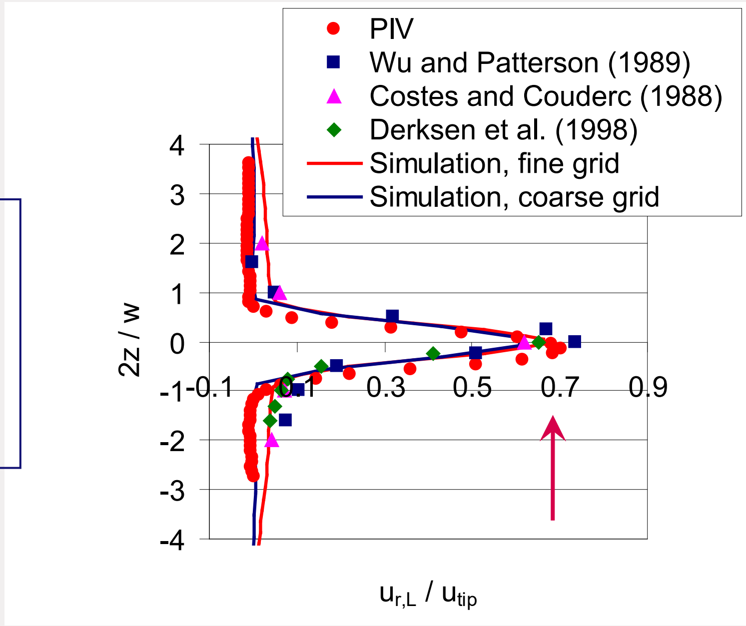
Results: stirred tank

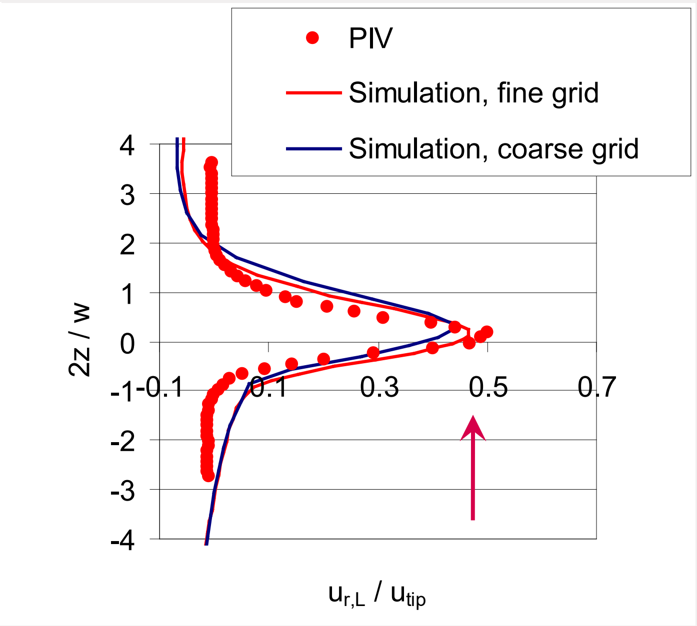




Results: stirred tank







Deen et al. (2002), Can. J. Chem. Eng. 80, 1-15

Wrap up

Turbulence modelling through: $\mu_t = C\rho\mathcal{V}\mathcal{L}$

- No model: Direct Numerical Simulations (DNS)
- Filtering: Large Eddy Simulations (LES)
- Averaging: *k*-*ε* model, RSM, ASM

RANS
100% modelled turbulence

LES
~90% resolved turbulence

DNS
100% resolved turbulence

Increasing computational cost and method accuracy

Extended Boussinesq approximation

$$\tau_{ij} = -\rho \overline{u_i' u_j'} = \mu_t \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right) = 2\mu_t E_{ij}$$

- **Only consider normal stresses (*i* = 1,2,3; *i* = *j*):**

$$\sum_i 2\mu_t E_{ii} = 2\mu_t \left[\frac{\partial U}{\partial x} + \frac{\partial V}{\partial y} + \frac{\partial W}{\partial z} \right] = 2\mu_t \,\mathrm{div}\,\mathbf{U} = 0$$

- **However:**

$$\sum_i \tau_{ii} = \sum_i -\rho \overline{u_i' u_i'} = -\rho \left(\overline{u'^2} + \overline{v'^2} + \overline{w'^2} \right) = -2\rho k$$

- **Define: (δ_{ij}= 1 if *i* = *j*) and (δ_{ij}= 0 if *i* ≠ *j*)**

$$\tau_{ij} = -\rho \overline{u_i' u_j'} = \mu_t \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right) - \frac{2}{3}\rho k \delta_{ij} = 2\mu_t E_{ij} - \frac{2}{3}\rho k \delta_{ij}$$